

# SEONGKU KANG

Assistant Professor, Department of Computer Science and Engineering, Korea University

Email: seongkukang@korea.ac.kr | [Google Scholar](#) | [Web Page](#) | [Lab Page](#)

## RESEARCH INTERESTS

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- Data science and machine learning for real-world applications
- Information retrieval and recommender systems

## EXPERIENCE

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| <b>Assistant Professor, Korea University</b> <ul style="list-style-type: none"><li>▪ Department of Computer Science and Engineering</li></ul>                                                                                                                 | Mar 2025 - Present    |
| <b>Postdoctoral Researcher, UIUC</b> <ul style="list-style-type: none"><li>▪ Data Mining Group (advisor: <a href="#">Prof. Jiawei Han</a>)</li></ul>                                                                                                          | Aug 2023 - Feb 2025   |
| <b>Research intern, Microsoft Research Asia</b> <ul style="list-style-type: none"><li>▪ Social Computing group (mentors: <a href="#">Dr. Xing Xie</a>, <a href="#">Dr. Jianxun Lian</a>)</li><li>· Award of excellence (Stars of Tomorrow, top 10%)</li></ul> | Oct 2022 - April 2023 |

## EDUCATION

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| <b>Ph.D., Computer Science and Engineering, POSTECH</b> <ul style="list-style-type: none"><li>▪ Data Intelligence Lab (advisor: <a href="#">Prof. Hwanjo Yu</a>)</li></ul> | Mar 2018 - Aug 2023 |
| <b>B.S., Computer Science and Engineering, NTU</b> <ul style="list-style-type: none"><li>▪ Exchange Student</li></ul>                                                      | Jul 2016 - May 2017 |
| <b>B.S., Computer Science and Engineering, Hanyang University</b> <ul style="list-style-type: none"><li>▪ Cumulative GPA: 4.43/4.50 (ranked 1st in CSE Dept.)</li></ul>    | Mar 2012 - Feb 2018 |

## PUBLICATIONS

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Unless otherwise specified, the paper is accepted as a long/regular paper. \* indicates equal contribution.

### Conferences

- CoRank: LLM-Based Compact Reranking with Document Features for Scientific Retrieval  
Runchu Tian, Xueqiang Xu, Bowen Jin, SeongKu Kang, Jiawei Han  
KDD'26 | ACM SIGKDD Conference on Knowledge Discovery and Data Mining
- SPRINT: Scalable and Predictive Intent Refinement for LLM-Enhanced Session-based Recommendation  
Gyuseok Lee, Wonbin Kweon, Z Yue, Y Liu, Y Liu, Susik Yoon, Dong Wang, [SeongKu Kang](#)  
SIGIR'26 | ACM SIGIR Conference on Research and Development in Information Retrieval
- Filling the Gaps: Selective Knowledge Augmentation for LLM Recommenders  
Jaehyun Lee, Sanghwan Jang, [SeongKu Kang](#)<sup>†</sup>, Hwanjo Yu<sup>†</sup>  
SIGIR'26 | ACM SIGIR Conference on Research and Development in Information Retrieval
- FLAME: Condensing Ensemble Diversity into a Single Network for Efficient Sequential Recommendation  
WooJoo Kim, Junyoung Kim, Jaehyung Lim, Seong Jin Choi, [SeongKu Kang](#)<sup>†</sup>, Hwanjo Yu<sup>†</sup>  
SIGIR'26 | ACM SIGIR Conference on Research and Development in Information Retrieval

5. MVIGER: Multi-View Variational Integration of Complementary Knowledge for Generative Recommender  
Tongyoung Kim, Soojin Yoon, SeongKu Kang, Jinyoung Yeo, Dongha Lee  
SIGIR'26 | ACM SIGIR Conference on Research and Development in Information Retrieval
6. Why These Documents? Explainable Generative Retrieval with Hierarchical Category Paths  
Sangam Lee, Ryang Heo, SeongKu Kang, Susik Yoon, Jinyoung Yeo, Dongha Lee  
ACL'26 Findings | Annual Meeting of the Association for Computational Linguistics
7. Continual Low-Rank Adapters for LLM-based Generative Recommender Systems  
Hyunsik Yoo, Ting-Wei Li, SeongKu Kang, Zhining Liu, Charlie Xu, Qilin Qi, Hanghang Tong  
ICLR'26 | International Conference on Learning Representations
8. Leveraging Historical and Current Interests for Continual Sequential Recommendation  
Gyuseok Lee, Hyunsik Yoo, Junyoung Hwang, SeongKu Kang<sup>†</sup>, Hwanjo Yu  
WSDM'26 | ACM International Conference on Web Search and Data Mining
9. PairSem: LLM-Guided Pairwise Semantic Matching for Scientific Document Retrieval  
Wonbin Kweon, Runchu Tian, SeongKu Kang, Pengcheng Jiang, Zhiyong Lu, Jiawei Han, Hwanjo Yu  
WWW'26 | ACM The Web Conference
10. CREAM: Continual Retrieval on Dynamic Streaming Corpora with Adaptive Soft Memory  
HuiJeong Son, Hyeongu Kang, Sunho Kim, Subeen Ho, SeongKu Kang, Dongha Lee, Susik Yoon  
KDD'26 | ACM SIGKDD Conference on Knowledge Discovery and Data Mining
11. Topic Coverage-based Demonstration Retrieval for In-Context Learning  
Wonbin Kweon, SeongKu Kang, Runchu Tian, Pengcheng Jiang, Jiawei Han, Hwanjo Yu  
EMNLP'25 | Conference on Empirical Methods in Natural Language Processing
12. Scientific Paper Retrieval with LLM-Guided Semantic-Based Ranking  
Yunyi Zhang, Ruozhen Yang, Siqi Jiao, SeongKu Kang, Jiawei Han  
EMNLP'25 Findings | Conference on Empirical Methods in Natural Language Processing
13. DeepRetrieval: Hacking Real Search Engines and Retrievers with Large Language Models via Reinforcement Learning  
Pengcheng Jiang, Jiacheng Lin, Lang Cao, Runchu Tian, SeongKu Kang, Zifeng Wang, Jimeng Sun, Jiawei Han  
COLM'25 | Conference on Language Modeling
14. Imagine All The Relevance: Scenario-Profiled Indexing with Knowledge Expansion for Dense Retrieval  
Sangam Lee, Ryang Heo, SeongKu Kang, Dongha Lee  
COLM'25 | Conference on Language Modeling
15. Delving into Instance-Dependent Label Noise in Graph Data: A Comprehensive Study and Benchmark  
Suyeon Kim, SeongKu Kang<sup>†</sup>, Dongwoo Kim, Jungseul Ok, Hwanjo Yu<sup>†</sup>  
KDD'25 | ACM SIGKDD Conference on Knowledge Discovery and Data Mining
16. Embracing Plasticity: Balancing Stability and Plasticity in Continual Recommender Systems  
Hyunsik Yoo, SeongKu Kang, Ruizhong Qiu, Charlie Xu, Fei Wang and Hanghang Tong  
SIGIR'25 | ACM SIGIR Conference on Research and Development in Information Retrieval
17. Personalized Preference Reasoning with Large Language Models for Accurate and Explainable Recommendation  
Jieyong Kim, Hyunseo Kim, Hyunjin Cho, SeongKu Kang, Buru Chang, Jinyoung Yeo and Dongha Lee  
SIGIR'25 | ACM SIGIR Conference on Research and Development in Information Retrieval

18. Uncertainty Quantification and Decomposition for LLM-based Recommendation  
Wonbin Kweon, Sanghwan Jang, SeongKu Kang<sup>†</sup>, Hwanjo Yu<sup>†</sup>  
WWW'25 | ACM The Web Conference
19. Chain-of-Factors Paper-Reviewer Matching  
Yu Zhang, Yanzhen Shen, SeongKu Kang, Xiusi Chen, Bowen Jin, Jiawei Han  
WWW'25 | ACM The Web Conference
20. Improving Scientific Document Retrieval with Concept Coverage-based Query Set Generation  
SeongKu Kang, Bowen Jin, Wonbin Kweon, Yu Zhang, Dongha Lee, Jiawei Han, Hwanjo Yu  
WSDM'25 | ACM International Conference on Web Search and Data Mining
21. Unsupervised Robust Cross-Lingual Entity Alignment via Neighbor Triple Matching with Entity and Relation Texts  
Soojin Yoon, Sungho Ko, Tongyoung Kim, SeongKu Kang, Jinyoung Yeo, Dongha Lee  
WSDM'25 | ACM International Conference on Web Search and Data Mining
22. Taxonomy-guided Semantic Indexing for Academic Paper Search  
SeongKu Kang, Yunyi Zhang, Pengcheng Jiang, Dongha Lee, Jiawei Han, Hwanjo Yu  
EMNLP'24 | Conference on Empirical Methods in Natural Language Processing
23. Continual Collaborative Distillation for Recommender System  
Gyuseok Lee\*, SeongKu Kang\*, Wonbin Kweon, Hwanjo Yu  
KDD'24 | ACM SIGKDD Conference on Knowledge Discovery and Data Mining
24. PEARL: A Review-driven Persona-Knowledge grounded Conversational Recommendation Dataset  
Minjin Kim\*, Minju Kim\*, Hana Kim, Beong-woo Kwak, Soyeon Jeon, Hyunseo Kim, SeongKu Kang, Youngjae Yu, Jinyoung Yeo, Dongha Lee  
ACL'24 Findings | Annual Meeting of the Association for Computational Linguistics
25. Self-Consistent Reasoning-based Aspect-Sentiment Quad Prediction with Extract-Then-Assign Strategy  
Jieyong Kim\*, Ryang Heo\*, Yongsik Seo, SeongKu Kang, Jinyoung Yeo, Dongha Lee  
ACL'24-short Findings | Annual Meeting of the Association for Computational Linguistics
26. Multi-Domain Sequential Recommendation via Domain Space Learning  
Junyoung Hwang, Hyunjun Ju, SeongKu Kang, Sanghwan Jang, Hwanjo Yu  
SIGIR'24 | ACM SIGIR Conference on Research and Development in Information Retrieval
27. Learning Discriminative Dynamics with Label Corruption for Noisy Label Detection  
Suyeon Kim, Dongha Lee, SeongKu Kang, Sukang Chae, Sanghwan Jang, Hwanjo Yu  
CVPR'24 | IEEE / CVF Computer Vision and Pattern Recognition
28. Improving Retrieval in Theme-specific Applications using a Corpus Topical Taxonomy  
SeongKu Kang, Shivam Agarwal, Bowen Jin, Dongha Lee, Hwanjo Yu, Jiawei Han  
WWW'24 | ACM The Web Conference
29. Top-Personalized-K Recommendation  
Wonbin Kweon, SeongKu Kang, Sanghwan Jang, Hwanjo Yu  
WWW'24 | ACM The Web Conference
30. Multi-Domain Recommendation to Attract Users via Domain Preference Modeling  
Hyunjun Ju, SeongKu Kang, Dongha Lee, Junyoung Hwang, Sanghwan Jang, Hwanjo Yu  
AAAI'24 | AAAI Conference on Artificial Intelligence
31. MvFS: Multi-view Feature Selection for Recommender System  
Youngjune Lee, Yeongjong Jeong, Keunchan Park, SeongKu Kang<sup>†</sup>  
CIKM'23-short | ACM International Conference on Information and Knowledge Management

32. Distillation from Heterogeneous Models for Top-K Recommendation  
SeongKu Kang, Wonbin Kweon, Dongha Lee, Jianxun Lian, Xing Xie, Hwanjo Yu  
WWW'23 | ACM The Web Conference
33. Learning Topology-Specific Experts for Molecular Property Prediction  
Suyeon Kim, Dongha Lee, SeongKu Kang, Seonghyeon Lee, Hwanjo Yu  
AAAI'23 | AAAI Conference on Artificial Intelligence
34. Consensus Learning from Heterogeneous Objectives for One-Class Collaborative Filtering  
SeongKu Kang, Dongha Lee, Wonbin Kweon, Junyoung Hwang, Hwanjo Yu  
WWW'22 | ACM The Web Conference
35. TaxoCom: Topic Taxonomy Completion with Hierarchical Discovery of Novel Topic Clusters  
Dongha Lee, Jiaming Shen, SeongKu Kang, Susik Yoon, Jiawei Han, Hwanjo Yu  
WWW'22 | ACM The Web Conference
36. Obtaining Calibrated Probabilities with Personalized Ranking Models  
Wonbin Kweon, SeongKu Kang, Hwanjo Yu  
AAAI'22 | AAAI Conference on Artificial Intelligence
37. Topology Distillation for Recommender System  
SeongKu Kang, Junyoung Hwang, Wonbin Kweon, Hwanjo Yu  
KDD'21 | ACM SIGKDD Conference on Knowledge Discovery and Data Mining
38. Bootstrapping User and Item Representations for One-Class Collaborative Filtering  
Dongha Lee, SeongKu Kang, Hyunjun Ju, Chanyoung Park, Hwanjo Yu  
SIGIR'21 | ACM SIGIR Conference on Research and Development in Information Retrieval
39. Unsupervised Proxy Selection for Session-based Recommender Systems  
Junsu Cho, SeongKu Kang, Dongmin Hyun, Hwanjo Yu  
SIGIR'21 | ACM SIGIR Conference on Research and Development in Information Retrieval
40. Learning Heterogeneous Temporal Patterns of User Preference for Timely Recommendation  
Junsu Cho, Dongmin Hyun, SeongKu Kang, Hwanjo Yu  
WWW'21 | ACM International World-Wide Web Conference
41. Bidirectional Distillation for Top-K Recommender System  
Wonbin Kweon, SeongKu Kang, Hwanjo Yu  
WWW'21 | ACM International World-Wide Web Conference
42. DE-RRD: A Knowledge Distillation Framework for Recommender System  
SeongKu Kang, Junyoung Hwang, Wonbin Kweon, Hwanjo Yu  
CIKM'20 | ACM International Conference on Information and Knowledge Management
43. Deep Rating Elicitation for New Users in Collaborative Filtering  
Wonbin Kweon, SeongKu Kang, Junyoung Hwang, Hwanjo Yu  
WWW'20-short | ACM International World-Wide Web Conference
44. Multi-Modal Component Embedding for Fake News Detection  
SeongKu Kang, Junyoung Hwang, Hwanjo Yu  
IMCOM'20 | IEEE International Conf. Ubiquitous Information Management and Communication
45. Semi-Supervised Learning for Cross-Domain Recommendation to Cold-start Users  
SeongKu Kang, Junyoung Hwang, Dongha Lee, Hwanjo Yu  
CIKM'19 | ACM International Conference on Information and Knowledge Management  
- ranked 13th among the most influential papers at CIKM 2019([link](#))

46. Densifying a Trust Network for Effective Collaborative Filtering  
SeongKu Kang, Jemin Wang, Yeon-Chang Lee, Sang-Wook Kim  
KDBC'17 | Korean DataBase Conference

## Journals

1. Improving Scientific Document Retrieval with Academic Concept Index  
Jeyun Lee, Junhyoung Lee, Wonbin Kweon, Bowen Jin, Yu Zhang, Susik Yoon, Dongha Lee, Hwanjo Yu, Jiawei Han, SeongKu Kang  
ACM Transactions on Intelligent Systems and Technology (SCIE) (To appear)
2. BPL: Bias-adaptive Preference Distillation Learning for Recommender System  
SeongKu Kang, Jianxun Lian, Dongha Lee, Wonbin Kweon, Sanghwan Jang, Jaehyun Lee, Jindong Wang, Xing Xie, Hwanjo Yu  
IEEE Transactions on Knowledge and Data Engineering (SCIE) (2026.01)
3. Unbiased, Effective, and Efficient Distillation from Heterogeneous Models for Recommender Systems  
SeongKu Kang, Wonbin Kweon, Dongha Lee, Jianxun Lian, Xing Xie, Hwanjo Yu  
ACM Transactions on Recommender Systems (2024.02)
4. Mitigating Viewpoint Sensitivity of Self-Supervised One-Class Classifiers  
Hyunjun Ju, Dongha Lee, SeongKu Kang, Hwanjo Yu  
Information Sciences (SCIE) (2022.09)
5. Personalized Knowledge Distillation for Recommender System  
SeongKu Kang, Dongha Lee, Wonbin Kweon, Hwanjo Yu  
Knowledge-Based Systems (SCIE) (2022.03)
6. Item-side Ranking Regularized Distillation for Recommender System  
SeongKu Kang, Junyoung Hwang, Wonbin Kweon, Hwanjo Yu  
Information Sciences (SCIE) (2021.11)

## AWARDS & FELLOWSHIPS

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### Awards

- Seoktop Lecture Award (Bigdata Analytics), Korea University, 2026
- Seoktop Lecture Award (Artificial Intelligence), Korea University, 2025
- Top-10 Paper Recognition, WSDM 2025 / ACM TIST Best Papers of WSDM 2025
- Outstanding reviewer, ACM KDD, 2024
- Stars of Tomorrow Award, Microsoft Research Asia, 2023
- POSTECH Research Award (4th place), National Research Foundation and POSTECH, Korea, 2022
- Music Playlist Recommendation Competition (3rd place), Kakao, 2020
- SIGIR Student Travel Award, ACM SIGIR, 2019-2020
- Dean's List, Hanyang University, 2018
- Best Paper Award (3rd place), Korean Database Conference (KDBC), 2017

### Fellowships

- POSTECHIAN Fellowship, POSTECH, 2023
- NAVER Ph.D Fellowship, NAVER, 2021
- NAVER Academic Scholarship, NAVER, 2017
- National Science and Technology Scholarship, Ministry of Education, Korea, 2012-2017
- Hanyang Brain Scholarship, Hanyang University, 2012

## PROFESSIONAL SERVICES

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### Program Committee/Reviewer

- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD): 2023 - 2026 (Area Chair: 2026)
- ACM International World-Wide Web Conference (WWW): 2024, 2025
- AAAI Conference on Artificial Intelligence (AAAI): 2024 - 2027
- ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR): 2025, 2026
- Annual Meeting of the Association for Computational Linguistics (ACL): 2025
- SIAM Conference on Data Mining (SDM): 2024
- Computer Vision and Pattern Recognition (CVPR): 2026
- Advances in Neural Information Processing Systems (NeurIPS): 2026
- ACM International Conference on Information and Knowledge Management (CIKM): 2026
- IEEE International Conference on Data Mining (ICDM): 2026
- ACM AI Summit: 2026

### Journal Reviewer

- Transactions on Information Systems: 2024, 2026
- Transactions on Knowledge and Data Engineering: 2024, 2025
- Knowledge-Based Systems: 2023, 2024
- Neural Computing and Applications: 2022

### Others

- Tutorial Organizer, DASFAA 2026
- Tutorial Organizer, CIKM 2025
- PhD Mentor, CIKM 2025 PhD Symposium

## TALKS

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|---------------------------------------------------------------------------------|----------|
| • CSE/GSAI Seminar, Pohang University of Science and Technology (POSTECH)       | Mar 2026 |
| • Tutorial: Continual Recommender Systems, ACM CIKM                             | Nov 2025 |
| • Panel discussion, Data Intelligence Workshop                                  | Jun 2025 |
| • CS/AI Seminar, Ulsan National Institute of Science and Technology (UNIST)     | Nov 2023 |
| • Invited Seminar, Electronic & Information Research Information Center (EIRIC) | Sep 2023 |
| • Invited Seminar, Social Computing Group, Microsoft Research Asia (MSRA)       | Jun 2023 |
| • Technology Showcase, Korea AI summit                                          | Dec 2022 |